

# Mathematics for Computer Science – Isomorphism, Homomorphism and Quotient Group.

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## 学习思路.

本章有一条联系紧密的知识链：从同态、同构出发，经过 Kernel、正规子群、商群等，最后到达同构定理。而这整条知识链围绕着一个核心目标：考察群结构。

# Isomorphisms(同构.)

## Motivation.

Many groups may have different appearances, however they are essentially same.

# Isomorphisms(同构).

## Definition of Isomorphism.

Two group  $(\mathbb{G}, \cdot)$  and  $(\mathbb{H}, \circ)$  are isomorphic if there exists a one-to-one and onto map  $\phi : \mathbb{G} \mapsto \mathbb{H}$  such that the group operation is preserved; that is,

$$\phi(a \cdot b) = \phi(a) \circ \phi(b)$$

for all  $a$  and  $b$  in  $\mathbb{G}$ . If  $\mathbb{G}$  is isomorphic to  $\mathbb{H}$ , we write  $\mathbb{G} \cong \mathbb{H}$ . The map  $\phi$  is called an isomorphism.

# Examples of Isomorphisms.

## Example

$\mathbb{Z}_4 \cong \langle i \rangle$ , since we can define a bijective map  $\phi : \mathbb{Z}_4 \mapsto \langle i \rangle$  by  $\phi(n) = i^n$ . The map  $\phi$  is one-to-one and onto, since

$$\phi(0) = 1$$

$$\phi(1) = i$$

$$\phi(2) = -1$$

$$\phi(3) = -i.$$

Moreover,  $\phi$  preserves the group operation, since

$$\phi(m+n) = i^{m+n} = i^m i^n = \phi(m)\phi(n).$$

# Examples of Isomorphisms.

## Isomorphic groups.

Since  $\mathbb{Z}_8^* = \{1, 3, 5, 7\}$ ,  $\mathbb{Z}_{12}^* = \{1, 5, 7, 11\}$ , we can find an isomorphism  $\phi$  to show that:

$$\mathbb{Z}_8^* \cong \mathbb{Z}_{12}^*$$

An isomorphism  $\phi : \mathbb{Z}_8^* \mapsto \mathbb{Z}_{12}^*$  is defined by :

$$1 \mapsto 1$$

$$3 \mapsto 5$$

$$5 \mapsto 7$$

$$7 \mapsto 11$$

Can you find another isomorphism between these two groups?

(Question.)

Is  $\mathbb{Z}_{61}^*$  isomorphic to  $\mathbb{Z}_{77}^*$ ? Why or why not?

# Examples of Isomorphisms.

Question-1.

Is  $\mathbb{Z}_n$  isomorphic to  $\mathbb{U}_n$ ? Why or why not?

# Examples of Isomorphisms.

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Question-2.

Is  $\mathbb{Z}_{61}^*$  isomorphic to  $\mathbb{Z}_{77}^*$ ? Why or why not?

# Theorem about Isomorphisms.

## Proposition

*Let  $\phi : \mathbb{G} \mapsto \mathbb{H}$  be an isomorphism of two groups, then the following statements are true.*

- 1  $\phi^{-1} : \mathbb{H} \mapsto \mathbb{G}$  is an isomorphism;
- 2  $|\mathbb{G}| = |\mathbb{H}|$ ;
- 3 If  $\mathbb{G}$  is abelian, then  $\mathbb{H}$  is abelian;
- 4 If  $\mathbb{G}$  is cyclic, then  $\mathbb{H}$  is cyclic;
- 5 if  $\mathbb{G}$  has a subgroup of order  $n$ , then  $\mathbb{H}$  has a subgroup of order  $n$ .

## Proof.

Left as an exercise.



# Theorem about Isomorphisms.

## Theorem

*All cyclic groups of infinite order are isomorphic to  $\mathbb{Z}$ .*

## Proof.

Suppose  $\mathbb{G}$  is a cyclic group with infinite order, and  $g \in \mathbb{G}$  is a generator. Define  $\phi : \mathbb{Z} \mapsto \mathbb{G}$  by  $\phi : n \mapsto g^n$ . Then

$$\phi(m+n) = g^{m+n} = g^m g^n = \phi(m)\phi(n).$$

Show  $\phi$  is a bijective map. Left as an exercise.



# Theorem about Isomorphisms.

## Theorem

*If  $\mathbb{G}$  is a cyclic group of order  $n$ , then  $\mathbb{G}$  is isomorphic to  $\mathbb{Z}_n$ .*

## Proof.

Let  $\mathbb{G}$  be a cyclic group with order  $n$ , generated by  $g$ . Define  $\phi : \mathbb{Z}_n \mapsto \mathbb{G}$  by  $\phi : k \mapsto g^k$ , where  $0 \leq k < n$ . Show  $\phi$  is an isomorphism. Left as an exercise. □

# Theorem about Isomorphisms.

## Corollary

*If  $\mathbb{G}$  is a cyclic group of order  $p$  where  $p$  is a prime, then  $\mathbb{G}$  is isomorphic to  $\mathbb{Z}_p$ .*

## Proof.

Easy!



## 单位根群.

$\mathbb{U}_n$  与  $\mathbb{Z}_n$  同构! 请问, 令  $n = p - 1$ ,  $\mathbb{U}_n$  与  $\mathbb{Z}_p^*$  同构吗? 这个答案揭示了什么信息?

# Theorem about Isomorphisms.

## Theorem

*The isomorphism of groups determines an equivalence relation on the class of all groups.*

## Proof.

Left as an exercise.



# Theorem about Isomorphisms.

## Theorem

*(Cayley) Every group is isomorphic to a group of permutations.*

## Proof.

Omitted. Note that, it is important.



# 关于 Isomorphisms 的提醒.

**提醒.**

同构是研究群结构的目标!

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提问.

同态的目标是什么?

# Homomorphisms. (同态)

## Definition of Homomorphism.

Two group  $(\mathbb{G}, \cdot)$  and  $(\mathbb{H}, \circ)$  are homomorphic if there exists a map  $\phi : \mathbb{G} \mapsto \mathbb{H}$  such that the group operation is preserved; that is,

$$\phi(a \cdot b) = \phi(a) \circ \phi(b)$$

for all  $a$  and  $b$  in  $\mathbb{G}$ . The map  $\phi$  is called a homomorphism.

## (Basic idea.)

We relax the requirement that an isomorphism of groups be bijective, we have a homomorphism.

# Examples of Homomorphisms.

## Example

$\mathbb{Z}$  是加法群，定义映射  $\phi: \mathbb{Z} \mapsto \mathbb{Z}$  为  $\phi(k) = 2k$ ,  $\forall k \in \mathbb{Z}$ 。可以验证  $\phi$  是一种群同态，因为

$$\phi(i+j) = 2(i+j) = 2i + 2j = \phi(i) + \phi(j)$$

$\phi$  把整数映射为偶数。偶数在加法下成群。

# Examples of Homomorphisms.

## Example of Homomorphisms.

Let  $\mathbb{G}$  be a group and  $g \in \mathbb{G}$ . Define a map  $\phi : \mathbb{Z} \mapsto \mathbb{G}$  by  $\phi(n) = g^n$ . Then  $\phi$  is a group homomorphism, since

$$\phi(m+n) = g^{m+n} = g^m g^n = \phi(m)\phi(n).$$

This homomorphism maps  $\mathbb{Z}$  onto the cyclic subgroup of  $\mathbb{G}$  generated by  $g$ .

# Examples of Homomorphisms.

## Example

设  $p$  为素数, 定义映射  $\phi: \mathbb{Z}_p^* \mapsto \mathbb{Z}_p^*$  为  $\phi(g) = g^2, \forall g \in \mathbb{Z}_p^*$ . 可验证,  $\phi$  是一种同态映射, 因为对任意的  $g_1, g_2 \in \mathbb{Z}_p^*$  满足:

$$\phi(g_1 g_2) = (g_1 g_2)^2 = g_1^2 g_2^2 = \phi(g_1) \phi(g_2).$$

# Normal subgroups (正规子群) .

## Definition of normal subgroups.

A subgroup  $\mathbb{H}$  of a group  $\mathbb{G}$  is normal in  $\mathbb{G}$  if  $g\mathbb{H} = \mathbb{H}g$  for all  $g \in \mathbb{G}$ .

## (Basic idea 1.)

A normal subgroup is a subgroup that the right cosets and the left cosets are precisely the same, and  $g\mathbb{H} = \mathbb{H}g$  represents a kind of "commutativity(交换性)".

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A normal subgroup is a subgroup that the right cosets and the left cosets are precisely the same, and  $g\mathbb{H} = \mathbb{H}g$  represents a kind of "commutativity(交换性)".

## (Basic idea 2.)

A subgroup  $\mathbb{H}$  of a group  $\mathbb{G}$  is normal in  $\mathbb{G}$  iff  $\forall g \in \mathbb{G}$ ,  $g\mathbb{H}g^{-1} \subseteq \mathbb{H}$ . Moreover, for all  $\forall g \in \mathbb{G}$ ,  $g\mathbb{H}g^{-1} = \mathbb{H}$

# Basic Properties of Normal Subgroup.

## Proposition

*Let  $\mathbb{G}$  be a group and  $\mathbb{H}$  be a subgroup of  $\mathbb{G}$ . Then the following statements are equivalent.*

- 1 The subgroup  $\mathbb{H}$  is a normal subgroup of  $\mathbb{G}$ , namely,  $g\mathbb{H} = \mathbb{H}g$  for all  $g \in \mathbb{G}$ .*
- 2 For all  $g \in \mathbb{G}$ ,  $g\mathbb{H}g^{-1} = \mathbb{H}$ .*

# Basic Properties of Normal Subgroup.

Proof.

(Proof of last proposition–1.) (1)  $\Rightarrow$  (2). Since  $\mathbb{H}$  is a normal subgroup of  $\mathbb{G}$ ,  $g\mathbb{H} = \mathbb{H}g$  for all  $g \in \mathbb{G}$ . Hence, for a given  $g \in \mathbb{G}$  and  $h \in \mathbb{H}$ , there exists an  $h' \in \mathbb{H}$  such that  $gh = h'g$ . Therefore,  $ghg^{-1} = h' \in \mathbb{H}$  or  $g\mathbb{H}g^{-1} \subseteq \mathbb{H}$ . For  $h \in \mathbb{H}$ ,  $g^{-1}hg = g^{-1}h(g^{-1})^{-1} \in \mathbb{H}$ . Hence,  $g^{-1}hg = h'$  for some  $h' \in \mathbb{H}$ . Therefore,  $h = gh'g^{-1} \in g\mathbb{H}g^{-1}$ , namely,  $\mathbb{H} \subseteq g\mathbb{H}g^{-1}$ .



# Basic Properties of Normal Subgroup.

Proof.

(Proof of last proposition-2.)

(2)  $\Rightarrow$  (1). Suppose that for all  $g \in \mathbb{G}$ ,  $g\mathbb{H}g^{-1} = \mathbb{H}$ . Then for any  $h \in \mathbb{H}$  there exists an  $h' \in \mathbb{H}$  such that  $ghg^{-1} = h'$ . Consequently,  $gh = h'g$  which means  $g\mathbb{H} \subseteq \mathbb{H}g$ . Similarly, we can prove that  $\mathbb{H}g \subseteq g\mathbb{H}$ . □

# Basic Properties of Homomorphisms.

## Proposition

*Proposition 1. Let  $\phi : \mathbb{G}_1 \mapsto \mathbb{G}_2$  be a homomorphism of groups. Then*

- ❶ *If  $e$  is the identity of  $\mathbb{G}_1$ , then  $\phi(e)$  is the identity of  $\mathbb{G}_2$ ;*
- ❷ *For any element  $g \in \mathbb{G}_1$ ,  $\phi(g^{-1}) = [\phi(g)]^{-1}$ ;*
- ❸ *If  $\mathbb{H}_1$  is a subgroup of  $\mathbb{G}_1$ , then  $\phi(\mathbb{H}_1)$  is a subgroup of  $\mathbb{G}_2$ ;*
- ❹ *If  $\mathbb{H}_2$  is a subgroup of  $\mathbb{G}_2$ , then  $\phi^{-1}(\mathbb{H}_2)$  is a subgroup of  $\mathbb{G}_1$ . Furthermore, if  $\mathbb{H}_2$  is normal in  $\mathbb{G}_2$ , then  $\phi^{-1}(\mathbb{H}_2)$  is normal in  $\mathbb{G}_1$ .*

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## Proof.

Omitted.



# Basic Properties of Homomorphisms.

## Definition

(Definition of Kernel.) Let  $\phi : \mathbb{G} \mapsto \mathbb{H}$  be a group homomorphism and  $e$  is the identity of  $\mathbb{H}$ . By previous proposition,  $\phi^{-1}(\{e\})$  is subgroup of  $\mathbb{G}$ . This subgroup is called the kernel of  $\phi$  and denoted by  $\text{Ker } \phi$ .

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## Example

设  $p$  为素数, 同态映射  $\phi : \mathbb{Z}_p^* \mapsto \mathbb{Z}_p^*$  定义为  $\phi(g) = g^2, \forall g \in \mathbb{Z}_p^*$ . 可验证,  $\phi$  把  $\{1, p-1\}$  映射为群  $\phi(\mathbb{Z}_p^*)$  的单位元  $1$ , 所以  $\text{Ker } \phi = \{1, p-1\}$ .

# Basic Properties of Homomorphisms.

## Proposition

*(Kernel.) Let  $\phi : \mathbb{G} \mapsto \mathbb{H}$  be a group homomorphism. Then the kernel of  $\phi$  is a normal subgroup of  $\mathbb{G}$ .*

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## Proof.

Trivial. Since the trivial subgroup of  $\mathbb{H}$  is normal.



## 关于 Homomorphisms 的提醒.

### 提醒

同态是研究群结构的重要手段：通过群同态去找正规子群，即同态映射的 Kernel.

## 习题.

- 如果  $\mathbb{H}_1$  和  $\mathbb{H}_2$  是群  $\mathbb{G}$  的正规子群, 证明  $\mathbb{H}_1\mathbb{H}_2$  也是群  $\mathbb{G}$  的正规子群。
- 证明,  $\mathbb{H}$  是  $\mathbb{G}$  的正规子群当且仅当对任意  $g \in \mathbb{G}$ ,  $g\mathbb{H}g^{-1} \subseteq \mathbb{H}$ 。
- 定义映射  $\phi: \mathbb{G} \mapsto \mathbb{G}$  为:  $g \mapsto g^2$ 。请证明  $\phi$  是一种群同态当且仅当  $\mathbb{G}$  是阿贝尔群。
- 设  $\phi: \mathbb{G} \mapsto \mathbb{H}$  是一种群同态。请证明: 如果  $\mathbb{G}$  是循环群, 则  $\phi(\mathbb{G})$  也是循环群; 如果  $\mathbb{G}$  是交换群, 则  $\phi(\mathbb{G})$  也是交换群。

# Quotient Groups(商群).

## Theorem

*If  $\mathbb{H}$  is a normal subgroup of a group  $\mathbb{G}$ , then the cosets of  $\mathbb{H}$  in  $\mathbb{G}$  form a group  $\mathbb{G}/\mathbb{H}$  under the operation  $(a\mathbb{H})(b\mathbb{H}) = ab\mathbb{H}$ . This group is call the quotient group or factor group of  $\mathbb{G}$  and  $\mathbb{H}$ .*

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## Understand the operation.

How to understand the operation  $(a\mathbb{H})(b\mathbb{H}) = ab\mathbb{H}$ ?

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## Understand the operation.

How to understand the operation  $(a\mathbb{H})(b\mathbb{H}) = ab\mathbb{H}$ ?

Since  $\mathbb{H}$  is normal, then:

$$(a\mathbb{H})(b\mathbb{H}) = (\mathbb{H}a)(b\mathbb{H}) = (ab\mathbb{H})\mathbb{H} = ab\mathbb{H}$$

## 进一步理解商群的群操作.

Understand the operation.

To understand the operation  $(a\mathbb{H})(b\mathbb{H}) = ab\mathbb{H}$ , we show that since  $\mathbb{H}$  is normal, then:

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使用了两种属性:

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- 结合律

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使用了两种属性:

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另一种理解.

有没有另一种理解方式? 借助正常的思维 (什么是正常思维?) 与底层定义 (直接借助群元素的操作!).

# 进一步理解商群的群操作.

## 正常思维!

我们需要证明  $(a\mathbb{H})(b\mathbb{H}) = ab\mathbb{H}$ , 即我们需要证明:

- $(a\mathbb{H})(b\mathbb{H}) \subseteq ab\mathbb{H}$
- $ab\mathbb{H} \subseteq (a\mathbb{H})(b\mathbb{H})$

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## 正常思维!

我们需要证明  $(a\mathbb{H})(b\mathbb{H}) = ab\mathbb{H}$ , 即我们需要证明:

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## Proof.

要证明  $(a\mathbb{H})(b\mathbb{H}) \subseteq ab\mathbb{H}$ , 即任取  $ah_1bh_2 \in a\mathbb{H}b\mathbb{H}$ , 证  $ah_1bh_2 \in ab\mathbb{H}$ . 但是, 这是容易的, 因为:

$$\begin{aligned} ah_1bh_2 &= h_3abh_2 \\ &= abh_4h_2 \\ &= abh_5 \in ab\mathbb{H} \end{aligned}$$

另一个方向, 证明  $ab\mathbb{H} \subseteq (a\mathbb{H})(b\mathbb{H})$  则留给大家作为练习。 □

# Theorem of Quotient Groups.

## Theorem

*(Quotient Groups). If  $\mathbb{H}$  is a normal subgroup of a group  $\mathbb{G}$ , then the cosets of  $\mathbb{H}$  in  $\mathbb{G}$  form a group  $\mathbb{G}/\mathbb{H}$  of order  $[\mathbb{G} : \mathbb{H}]$ .*

# Theorem of Quotient Groups.

Proof.

(Basic ideas.)

- 1 What is the group operation?  $(a\mathbb{H})(b\mathbb{H}) = ab\mathbb{H}$

# Theorem of Quotient Groups.

## Proof.

(Basic ideas.)

- 1 What is the group operation?  $(a\mathbb{H})(b\mathbb{H}) = ab\mathbb{H}$
- 2 Prove this operation is well-defined, i.e., the group operation must be independent of the choice of coset representative.  
Let  $a\mathbb{H} = b\mathbb{H}$ ,  $c\mathbb{H} = d\mathbb{H}$ . We must prove that

$$(a\mathbb{H})(c\mathbb{H}) = ac\mathbb{H} = bd\mathbb{H} = (b\mathbb{H})(d\mathbb{H})$$

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- 3 Why we need "well-defined"?

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- 3 Why we need "well-defined"?
- 4 Check the axioms of group. Easy!



# Theorem of Quotient Groups.

## Remark

(良定义操作.) 所谓良定义的操作, 就是要求操作独立于所参与操作的代表元。比如, 对任意群  $\mathbb{G}$  和其上的某种操作  $\psi : \mathbb{G} \mapsto \mathbb{G}$ , 要求  $\psi$  良定义就是要求对任意的群元  $a, b \in \mathbb{G}$ , 如果  $a = b$ , 则  $\psi(a) = \psi(b)$ 。一眼看上去, 这个要求很无理, 毫无意义, 但是对于商群来说就必不可少。请注意, 商群中操作的是陪集,  $a\mathbb{H} = b\mathbb{H}$  并不意味  $a = b$ 。

# Theorem of Quotient Groups.

## Remark

(群操作的良定义性.) Let  $a\mathbb{H} = b\mathbb{H}$ ,  $c\mathbb{H} = d\mathbb{H}$ . We must prove that

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$$(a\mathbb{H})(c\mathbb{H}) = ac\mathbb{H} = bd\mathbb{H} = (b\mathbb{H})(d\mathbb{H})$$

For  $a = bn_1$  and  $c = dn_2$  for some  $n_1$  and  $n_2$  in  $\mathbb{H}$ . Hence,

$$\begin{aligned}ac\mathbb{H} &= bn_1dn_2\mathbb{H} \\&= bn_1d\mathbb{H} \\&= bn_1\mathbb{H}d \\&= b\mathbb{H}d \\&= bd\mathbb{H}\end{aligned}$$

## 小结-商群的构造.

### (构造商群的思路).

- 首先, 给定群  $G$ , 找到它的一个正规子群  $H$ , 比如通过同态映射的 Kernel 来找到一个正规子群。
- 其次, 构造商群的元素, 即通过  $H$  构造陪集。
- 第三, 定义商群的群操作, 即陪集的乘法, 并验证这个操作的良好定义性。
- 最后, 验证在所定义的乘法上, 商群的确实满足群公理: 结合律、单位元、逆元、封闭性。

## Example for Quotient Groups.

(Example of Quotient Groups).

Consider the normal subgroup  $3\mathbb{Z}$  of  $\mathbb{Z}$ . The cosets of  $3\mathbb{Z}$  in  $\mathbb{Z}$  are

$$0 + 3\mathbb{Z} = \{\dots, -3, 0, 3, 6, \dots\}$$

$$1 + 3\mathbb{Z} = \{\dots, -2, 1, 4, 7, \dots\}$$

$$2 + 3\mathbb{Z} = \{\dots, -1, 2, 5, 8, \dots\}.$$

The group  $\mathbb{Z}/3\mathbb{Z}$  is given by the multiplicative table below.

| +                 | $0 + 3\mathbb{Z}$ | $1 + 3\mathbb{Z}$ | $2 + 3\mathbb{Z}$ |
|-------------------|-------------------|-------------------|-------------------|
| $0 + 3\mathbb{Z}$ | $0 + 3\mathbb{Z}$ | $1 + 3\mathbb{Z}$ | $2 + 3\mathbb{Z}$ |
| $1 + 3\mathbb{Z}$ | $1 + 3\mathbb{Z}$ | $2 + 3\mathbb{Z}$ | $0 + 3\mathbb{Z}$ |
| $2 + 3\mathbb{Z}$ | $2 + 3\mathbb{Z}$ | $0 + 3\mathbb{Z}$ | $1 + 3\mathbb{Z}$ |

## Example for Quotient Groups.

### (Quotient Groups of $\mathbb{Z}_n^*$ ).

Let  $n = 15$ , then  $\mathbb{Z}_n^* = \{1, 2, 4, 7, 8, 11, 13, 14\}$ . Let  $g = 2$ , we set  $\mathbb{S} = \langle g \rangle = \{1, 2, 4, 8\}$  which is a subgroup of  $\mathbb{Z}_n^*$ . Then  $\mathbb{Z}_n^*/7\mathbb{S} = \{\mathbb{S}, 7\mathbb{S}\}$ , please check that  $\mathbb{S}$  is the identity,  $7\mathbb{S}$ 's inverse is itself, namely  $(7\mathbb{S})(7\mathbb{S}) = 4\mathbb{S} = \mathbb{S}$ .

# Theorem of Quotient Groups.

## Example

(非良定义操作与商群) 设有限循环群  $G = \langle g \rangle$ ,  $H$  是群  $G$  的子群, 自然也是正规子群, 于是形成商群  $G/H$ 。定义映射  $\phi_g : G/H \mapsto \mathbb{Z}$ , 对任意的  $aH$ , 如果  $a = g^k$ , 则  $\phi_g(aH) = k$ 。可以验证这是一种非良定义映射。

# Theorem of Quotient Groups.

## Example

(非良定义操作与商群) 比如, 设  $p = 7$ , 则  $\mathbb{Z}_p^* = \{1, 2, 3, 4, 5, 6\}$ , 取生成元  $g = 3$ 。记  $S = \{1, 6\}$ , 可验证  $S$  是  $\mathbb{Z}_p^*$  的子群。所以形成商群,  $\mathbb{Z}_p^*/S = \{S, 2S, 3S\}$ 。此时  $3S = 4S$ ,  $\phi_3(3S) = 1$ , 但是  $\phi_3(4S) = 4$ 。也就是说, 对同一个陪集, 因为选取了不同的代表元, 得到的映射值不同, 该操作不能独立于代表元的选取。

## 思路.

商群怎么用？答：以上知识点可以形成这样一条知识链：从一个群同态  $\phi: \mathbb{G}_1 \mapsto \mathbb{G}_2$  出发。首先可以找到  $\phi$  的 Kernel，于是得到一个  $\mathbb{G}_1$  的正规子群；接着，构造  $\mathbb{G}_1$  上的商群  $\mathbb{G}_1 / \text{Ker } \phi$ 。再通过“同构定理”完成一直以来的主要任务：研究群（商群）的结构！)

# Canonical Homomorphism.

## 正规同态（自然同态）.

Let  $\mathbb{H}$  be a normal subgroup of  $\mathbb{G}$ , define a map

$$\phi : \mathbb{G} \mapsto \mathbb{G}/\mathbb{H}$$

by

$$\phi(g) = g\mathbb{H}.$$

This map is indeed a homomorphism, check it! We call this map a natural or canonical homomorphism, and  $\text{Ker } \phi = \mathbb{H}$ .

# First Isomorphism Theorem.

## Theorem

*(First Isomorphism Theorem.) If  $\psi : \mathbb{G} \mapsto \mathbb{H}$  is a group homomorphism with  $\mathbb{K} = \text{Ker } \psi$ , then  $\mathbb{K}$  is normal in  $\mathbb{G}$ . Let  $\phi : \mathbb{G} \mapsto \mathbb{G}/\mathbb{K}$  be the canonical homomorphism. Then there exists a unique isomorphism  $\eta : \mathbb{G}/\mathbb{K} \mapsto \psi(\mathbb{G})$  such that  $\psi = \eta\phi$ .*

# First Isomorphism Theorem.

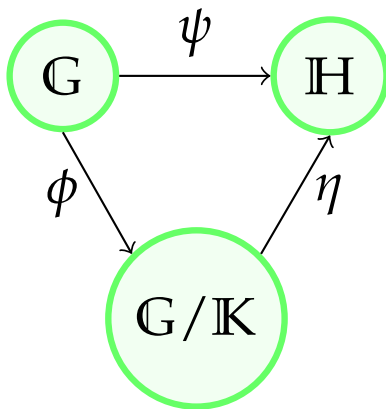


Figure: A diagrammatic interpretation of First Isomorphism Theorem.

# 关于第一同构定理的直观认识.

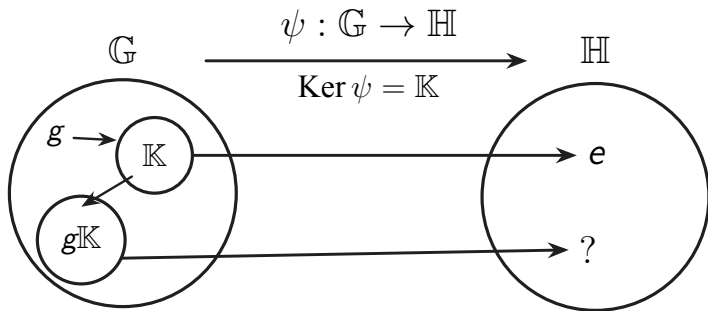


Figure: 关于第一同构定理的直观认识.

# First Isomorphism Theorem.

(证明思路.)

- 1 Define  $\eta : \mathbb{G}/\mathbb{K} \mapsto \psi(\mathbb{G})$  by  $\eta(g\mathbb{K}) = \psi(g)$ ;
- 2 Prove  $\eta$  is well-defined;
- 3 Prove that  $\eta$  is a homomorphism and is a bijective map.

# First Isomorphism Theorem.

(证明第一步.)

- 定义映射  $\eta : \mathbb{G}/\mathbb{K} \mapsto \psi(\mathbb{G})$  为  $\eta(g\mathbb{K}) = \psi(g)$ 。

# First Isomorphism Theorem.

## (证明第二步.)

- 证明  $\eta$  是良定义映射, 即证明如果对于两个不同的  $g_1, g_2 \in \mathbb{G}$ , 有  $g_1\mathbb{K} = g_2\mathbb{K}$ , 则  $\eta(g_1\mathbb{K}) = \eta(g_2\mathbb{K})$ 。因为  $g_1\mathbb{K} = g_2\mathbb{K}$ , 则存在  $k \in \mathbb{K}$  使得  $g_1 = g_2k$ , 这仅仅利用了陪集相等的定义。因此:

$$\eta(g_1\mathbb{K}) = \psi(g_1) = \psi(g_2k) = \psi(g_2)\psi(k) = \psi(g_2) = \eta(g_2\mathbb{K})$$

请注意, 上式中第四个等号之所以成立是因为  $k \in \mathbb{K}$ , 而  $\mathbb{K}$  是  $\psi$  的 Kernel,  $k$  必然映射到  $\mathbb{H}$  的单位元。这就证明了  $\eta$  映射独立于陪集代表元的选择。之所以  $\eta$  是唯一定义的, 因为给定了  $\psi$  和  $\phi$ , 而  $\psi = \eta\phi$ , 且  $\phi$  是满射。

# First Isomorphism Theorem.

(一个问题.)

确保  $\eta$  的唯一性, 为什么需要  $\phi$  是满射? (课后练习!)

## (证明第三步.)

- 可证明  $\eta$  是同态且是双射。 $\eta$  是同态，因为：

$$\begin{aligned}\eta(g_1 \mathbb{K} g_2 \mathbb{K}) &= \eta(g_1 g_2 \mathbb{K}) \\ &= \psi(g_1 g_2) \\ &= \psi(g_1) \psi(g_2) \\ &= \eta(g_1 \mathbb{K}) \eta(g_2 \mathbb{K})\end{aligned}$$

$\eta$  显然是满射，最后，只需证明  $\eta$  是单射。

# First Isomorphism Theorem.

## (证明第三步.)

- 可证明  $\eta$  是同态且是双射。  $\eta$  是同态， 因为：

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$\eta$  显然是满射， 最后， 只需证明  $\eta$  是单射。

## (一点点提醒.)

为什么  $\eta$  显然是满射？

# First Isomorphism Theorem.

## (证明第三步.)

- 证明  $\eta$  是单射。任取  $g_1\mathbb{K}, g_2\mathbb{K}$ , 假设  $\eta(g_1\mathbb{K}) = \eta(g_2\mathbb{K})$ , 则  $\psi(g_1) = \psi(g_2)$ 。  $g_1$  和  $g_2$  是  $\mathbb{G}$  中两个不同的元素, 所以

$$\psi(g_1) = \psi(g_2 g_2^{-1} g_1) = \psi(g_2) \psi(g_2^{-1} g_1)$$

即  $\psi(g_2^{-1} g_1)$  是  $\psi(\mathbb{G})$  的单位元, 也即  $g_2^{-1} g_1 \in \mathbb{K}$ 。所以  $g_2^{-1} g_1 \mathbb{K} = \mathbb{K}$ , 最后得到  $g_1 \mathbb{K} = g_2 \mathbb{K}$ 。

# Example for First Isomorphism Theorem.

## (Homomorphism from Cyclic Group.)

设  $\mathbb{G}$  是由生成元  $g$  生成的循环群。定义映射  $\phi: \mathbb{Z} \mapsto \mathbb{G}$  为  $n \mapsto g^n$ ,  $\forall n \in \mathbb{Z}$ 。  $\phi$  是同态映射, 因为:

$$\phi(m+n) = g^{m+n} = g^m g^n = \phi(m)\phi(n)。$$

$\phi$  显然是满射。如果  $\mathbb{G}$  的阶为  $m$ , 因为  $g$  是生成元, 则  $\text{ord}(g) = m$ 。于是,  $g^m = e$ , 且有  $\text{Ker } \phi = m\mathbb{Z}$ 。根据第一同构定理, 则有:

$$\mathbb{Z} / \text{Ker } \phi = \mathbb{Z} / m\mathbb{Z} \cong \mathbb{G}。$$

如果  $\mathbb{G}$  是无限阶, 则  $g$  也是无限阶, 则  $\text{Ker } \phi = \{0\}$ , 则  $\mathbb{Z}$  与  $\mathbb{G}$  同构。因此, 两个循环群同构当且仅当它们有相同的阶。在同构的意义上, 只有两种循环群:  $\mathbb{Z}$  和  $\mathbb{Z}_n$ 。

## Example for First Isomorphism Theorem.

(Homomorphism from  $\mathbb{Z}_p^*$  to  $\mathbb{Z}_p^*$ .)

Let  $p$  be a prime,  $\mathbb{Z}_p^*$  is a cyclic group. Define a map  $\phi : \mathbb{Z}_p^* \mapsto \mathbb{Z}_p^*$  by  $\phi(g) = g^2$  for all  $g \in \mathbb{Z}_p^*$ . Then  $\phi$  is a group homomorphism, since

$$\phi(g_1 g_2) = (g_1 g_2)^2 = g_1^2 g_2^2 = \phi(g_1) \phi(g_2).$$

Clearly  $\phi$  is not onto, and  $\text{Ker } \phi = \{1, p-1\}$  is a normal subgroup of  $\mathbb{Z}_p^*$ . We know  $\text{Ker } \phi$  because we believe that the following equation

$$x^2 \equiv 1 \pmod{p}$$

has only two solutions, namely 1 and  $p-1$ . Check that  $\mathbb{S} = \{\phi(g) : \text{for all } g \in \mathbb{Z}_p^*\}$  is a group. What is the order of  $\mathbb{S}$ ? By the First Isomorphism Theorem,  $|\mathbb{S}| = |\mathbb{Z}_p^* / \text{Ker } \phi| = |\mathbb{Z}_p^*| / |\text{Ker } \phi|$ .

## Example for First Isomorphism Theorem.

(Homomorphism from  $\mathbb{Z}_n^*$  to  $\mathbb{Z}_n^*$ .)

Let  $n = pq$  be a composite integer,  $p$  and  $q$  are two primes, and  $\mathbb{Z}_n^*$  is a group. Define a map  $\phi : \mathbb{Z}_n^* \mapsto \mathbb{Z}_n^*$  by  $\phi(g) = g^2$  for all  $g \in \mathbb{Z}_n^*$ . Then  $\phi$  is a group homomorphism.  $S = \{\phi(g) : \text{for all } g \in \mathbb{Z}_n^*\}$ , if we know the order of  $\text{Ker } \phi$ , then we know the order of  $S = |\mathbb{Z}_n^*|/|\text{Ker } \phi|$  by the First Isomorphism Theorem. How many solutions does the following equation have?

$$x^2 \equiv 1 \pmod{n}$$

Unfortunately, we don't solve it until we learn CRT.

# 一个值得记忆的结论.

一个好用的结论.

设  $\phi: \mathbb{G} \mapsto \mathbb{H}$  是一种群同态。 $\phi$  是单射当且仅当  $\text{Ker } \phi = \{e\}$

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## 证明.

- $\Rightarrow$ :  $\phi$  是单射, 显然  $\text{Ker } \phi = \{e\}$ 。
- $\Leftarrow$ :  $\text{Ker } \phi = \{e\}$ , 根据第一同构定理, 有  $\mathbb{G} \cong \mathbb{G} / \text{Ker } \phi \cong \phi(\mathbb{G})$ 。所以  $\phi$  是单射。

## 例子-1.

同构.

设  $G$  和  $H$  是群。证明  $G \times H$  同构于  $H \times G$

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证明.

- 思路：构造第一同构定理的“三角关系”。并使得  $\text{Ker } \phi = \{e\}$ 。

## 例子-2.

### 同构.

设  $H$  和  $K$  是群  $G$  的正规子群, 且  $H \cap K = \{e\}$ . 证明  $G$  同构于  $G/H \times G/K$  的一个子群。

## 例子-2.

### 同构.

设  $H$  和  $K$  是群  $G$  的正规子群, 且  $H \cap K = \{e\}$ 。证明  $G$  同构于  $G/H \times G/K$  的一个子群。

### 证明.

- 思路: 构造第一同构定理的“三角关系”。并使得  $\text{Ker } \phi = \{e\}$ 。

## Second Isomorphism Theorem.

### Theorem

(第二同构定理.)  $H$  是群  $G$  的子群 (不必然是正规子群),  $K$  是群  $G$  的正规子群。则  $HK$  是群  $G$  的子群,  $H \cap K$  是  $H$  的正规子群, 且

$$H/(H \cap K) \cong HK/K.$$

# Correspondence Theorem.

## Correspondence Theorem. (对应定理)

Let  $\mathbb{H}$  be a normal subgroup of a group  $\mathbb{G}$ . Then  $\mathbb{H} \mapsto \mathbb{H}/\mathbb{H}$  is a one-to-one correspondence between the set of subgroups  $\mathbb{H}$  containing  $\mathbb{H}$  and the set of subgroups of  $\mathbb{G}/\mathbb{H}$ . Furthermore, the normal subgroups of  $\mathbb{G}$  containing  $\mathbb{H}$  correspond to normal subgroups of  $\mathbb{G}/\mathbb{H}$ .

# Correspondence Theorem.(对应定理)

## Understanding Correspondence Theorem.

- 1 What is the map  $\mathbb{H} \mapsto \mathbb{H}/\mathbb{H}$ ?

# Correspondence Theorem.(对应定理)

## Understanding Correspondence Theorem.

- ① What is the map  $\mathbb{H} \mapsto \mathbb{H}/\mathbb{H}$ ?
- ② A map:  $\{\text{the set of subgroups } \mathbb{H} \text{ containing } \mathbb{H}\} \mapsto \{\text{the set of subgroups of } \mathbb{G}/\mathbb{H}\}$

# Correspondence Theorem.(对应定理)

## Understanding Correspondence Theorem.

- 1 What is the map  $\mathbb{H} \mapsto \mathbb{H}/\mathbb{H}$ ?
- 2 A map:  $\{\text{the set of subgroups } \mathbb{H} \text{ containing } \mathbb{H}\} \mapsto \{\text{the set of subgroups of } \mathbb{G}/\mathbb{H}\}$
- 3 To understand what is a subgroup of  $\mathbb{G}/\mathbb{H}$ ?

# Correspondence Theorem.

## Proof ideas of the Correspondence Theorem.

- 1  $\mathbb{H}/\mathbb{H}$  is a subgroup of  $\mathbb{G}/\mathbb{H}$ ;
- 2 The map  $\mathbb{H} \mapsto \mathbb{H}/\mathbb{H}$  is one-to-one and onto;
- 3  $\mathbb{H}$  is normal in  $\mathbb{G}$ , if and only if  $\mathbb{H}/\mathbb{H}$  is normal in  $\mathbb{G}/\mathbb{H}$ .

# Third Isomorphism Theorem.

## Theorem

(第三同构定理)  $H$  和  $K$  是群  $G$  的正规子群, 且  $K \subseteq H$ 。则:

$$G/H \cong \frac{G/K}{H/K}.$$

## 习题.

- 证明：如果  $H$  是群  $G$  上指标为 2 的子群，则  $H$  是  $G$  的正规子群。
- 给定任意群  $G$ ， $H$  是群  $G$  的正规子群。请证明，如果群  $G$  是阿贝尔群，则商群  $G/H$  也是阿贝尔群。
- 给定任意群  $G$ ， $H$  是群  $G$  的正规子群。请证明，如果群  $G$  是循环群，则商群  $G/H$  也是循环群。
- 设  $p$  和  $q$  是两个不同的素数， $G$  是阶为  $pq$  的阿贝尔群，请证明  $G$  是循环群。

## 习题.

设  $p$  和  $q$  是两个不同的素数,  $\mathbb{G}$  是阶为  $pq$  的阿贝尔群, 请证明  $\mathbb{G}$  是循环群。

思路: 找阶为  $p$  的群元  $g_1$  和阶为  $q$  的群元  $g_2$ , 则可知  $g_1g_2$  的阶是  $pq$ , 则  $\mathbb{G}$  是循环群。

# 习题讲解.

## 习题.

设  $p$  和  $q$  是两个不同的素数,  $\mathbb{G}$  是阶为  $pq$  的阿贝尔群, 请证明  $\mathbb{G}$  是循环群。

思路: 找阶为  $p$  的群元  $g_1$  和阶为  $q$  的群元  $g_2$ , 则可知  $g_1g_2$  的阶是  $pq$ , 则  $\mathbb{G}$  是循环群。

## Proof.

不妨假设存在阶为  $p$  的群元  $g_1$ ,  $\mathbb{H} = \langle g_1 \rangle$  为循环群。可得阶为  $q$  的商群  $\mathbb{G}/\mathbb{H}$ , 它也必然为循环群。记  $\mathbb{G}/\mathbb{H}$  的生成元是  $g_2\mathbb{H}$ 。此时,  $g_2$  的阶不能是 1。同时,  $g_2$  的阶不能是  $p$ , 否则  $(g_2\mathbb{H})^p = \mathbb{H}$ 。如果  $p < q$ , 则  $g_2\mathbb{H}$  不能是生成元。如果  $p > q$ , 则  $q$  整除  $p$ , 这是不可能的! 所以,  $g_2$  的阶只能是  $q$  或者  $pq$ , 无论是什么情况,  $\mathbb{G}$  都是循环群。□

# 关于第一同构定理运用的习题.

## 习题.

- 证明:  $\mathrm{GL}_n(R)/\mathrm{SL}_n(R)$  与乘法群  $\mathbb{R}^*$  同构。
- 证明商群  $\mathbb{R}/\mathbb{Z}$  同构于圆圈群  $\mathbb{O}$ 。
- 设  $\phi: \mathbb{G} \mapsto \mathbb{H}$  是一种群同态。请证明  $\phi$  是单射当且仅当  $\mathrm{Ker} \phi = \{e\}$ 。
- 给定有限阿贝尔群  $\mathbb{G}$  和正整数  $n$ , 定义映射  $\phi: \mathbb{G} \mapsto \mathbb{G}$  为,  $\forall g \in \mathbb{G}, \phi(g) = g^n$ 。请分析并证明, 在什么条件下  $\phi$  会是一种单射?